

```

> use BookFlow
< switched to db BookFlow
> db.book_metadata.insertMany([
  {
    title: "The Silent Algorithm",
    author: "R. Menon",
    genre: "Technology",
    publishedYear: 2022,
    formats: ["Hardcover", "Digital", "Audiobook"],
    specs: { pages: 340, fileSize_MB: 4.2, drm: true, narrator: "K. Rao", runtime_min: 610 },
    avgRating: 4.6,
    reviews: [
      { member_id: "M1001", rating: 5, comment: "Excellent deep dive into digital systems design.", date: ISODate("2023-01-15") },
      { member_id: "M1042", rating: 4, comment: "Good Digital edition.", date: ISODate("2023-02-20") }
    ]
  },
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  }
])

```

```

< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a912f3ef4f77b8803089ee5'),
    '1': ObjectId('6a912f3ef4f77b8803089ee6'),
    '2': ObjectId('6a912f3ef4f77b8803089ee7'),
    '3': ObjectId('6a912f3ef4f77b8803089ee8')
  }
}
BookFlow>

```

```
> db.book_metadata.find().pretty()
< {
  _id: ObjectId('6a912f3ef4f77b8803089ee5'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
```

```
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
```

BookFlow> |

```
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
```

BookFlow>





Welcome

> mongosh: Geya



>_MONGOSH

```
> db.book_metadata.find({
  formats: { $regex: "Digital", $options: "i" }
})
< {
  _id: ObjectId('6a912f3ef4f77b8803089ee5'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
    {
      member_id: 'M1042'
```

MongoDB Compass - Geya/Shell

Connections Edit View Help

Compass

{ } My Queries

Data Modeling

CONNECTIONS (1)

Search connections

▼ Geya

- Week_3
 - db
- admin
- config
- local
- studentdb
- test_db

»_MONGOSH

```
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } },
  { title: 1, "reviews.$": 1 }
)
< {
  _id: ObjectId('6a912f3ef4f77b8803089ee5'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a912f3ef4f77b8803089ee6'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read.',
      date: 2022-05-10T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a912f3ef4f77b8803089ee8'),
  title: 'Quantum Horizons',
```

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»_MONGOSH

```
> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: "$reviews" },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  { $sort: { avgReviewRating: -1 } },
  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: { $round: ["$avgReviewRating", 2] },
      totalReviews: 1
    }
  }
])
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
```

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mongosh: Geya

```
>_MONGOSH
{ $sort: { avgReviewRating: -1 } },

{
  $project: {
    _id: 0,
    title: "$_id",
    author: 1,
    avgReviewRating: { $round: [ "$avgReviewRating", 2 ] },
    totalReviews: 1
  }
}
})
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
}
BookFlow>
```

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mongosh: Geya

```
>_MONGOSH
> db.book_metadata.createIndex({
  "reviews.comment": "text"
})
< reviews.comment_text
> db.book_metadata.find({
  $text: { $search: "Digital" }
})
< {
  _id: ObjectId('6a912f3ef4f77b8803089ee5'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
    }
  ]
}
```


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»_MONGOSH

```
> db.book_metadata.find({
  $text: { $search: "Digital" }
}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExploreReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'BookFlow.book_metadata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
```

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Data Modeling

CONNECTIONS (1)

Search connections

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 - db
- admin
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- studentdb
- test_db

»_MONGOSH

```
> db.book_metadata.createIndex({ formats: 1 })
< formats_1
> db.book_metadata.createIndex({ avgRating: -1 })
< avgRating_-1
> db.book_metadata.find({
  avgRating: { $gt: 4 }
}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      avgRating: {
        '$gt': 4
      }
    },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: '6A4442CC',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExploreReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'BookFlow.book_metadata',
      inputStage: {
        stage: 'IXSCAN',
```

